Practice Final Exam (F25)

Started: Dec 13 at 6:07pm

Quiz Instructions

Question 1 1 pts

Refer to the BoosterPack pinout diagram and answer the following questions:

1. Which port corresponds to the **blue** BoosterPack LED?
2. Which pin is the **blue** BoosterPack LED connected to?
3. Which Timer_A module is used by the **blue** BoosterPack LED?
4. Which Timer_A channel is the **blue** BoosterPack LED connected to?

For each of your answers, use single-digit, base-10 numbers only. Do not use any hexadecimal or binary numbers, and do not use macros for any answer.

Question 2 1 pts

A Timer_A uses the system clock of 48 MHz as its source clock with a clock divider of 14. Assume the Timer_A is using the **maximum possible load value**, is in count-up mode, and is using the Toggle output mode. How much time elapses between **two rollovers** for this Timer_A? ***Give your answer as an integer in microseconds, rounded up to the closest integer.***

Question 3 1 pts

Consider a Timer_A in the **count-up** mode with CCR0 = 25,000 and CCR1 = 4,000. Assume the system clock is 48 MHz and the clock divider is set to 48. Fill the below table for the PWM generated using this setup.

Output mode	PWM Frequency (Hz) (round up with no decimal points)	Duty cycle percent (A number between 0 and 100 - round up with no decimal points)

Set/Reset		
Reset/Set		

Question 4 1 pts

Suppose we want to use the buzzer on the microcontroller. We want the buzzer to play the note F5 (659.25 Hz). Assume that the system clock frequency is 48 MHz and the output mode is configured to Set/Reset.

1. What is the ***minimum*** prescaler value we can use? [ ]
2. What should the corresponding load value be? [ ]
3. What duty cycle percentage should we use? [ ] (%)

Give your answer as an integer rounded up with no decimal points for load value , and between 0-100 for duty.

Question 5 1 pts

Consider a processor whose nominal operating voltage is 5.0V. We also know the processor is prone to making mistakes when operating below 3.8V. Which one(s) of the below threshold(s) is suitable for implementing brownout reset?

1. One threshold at 3.9V [Select]
2. One threshold at 3.8V [Select]
3. One threshold at 3.7V [Select]
4. Two thresholds: 3.84V and 3.89V [Select]
5. Two thresholds: 3.76V and 3.88V [Select]
6. Two thresholds: 3.93V and 4.02V [Select]

Question 6 1 pts

We are running a program on a microcontroller that has four enabled interrupts: Interrupt #1, Interrupt #2, Interrupt #3, and Interrupt #4. Each interrupt has its own, corresponding ISR. They have the following priority (where a higher number means higher priority):

Interrupt	Priority
Interrupt #1	2
Interrupt #2	3
Interrupt #3	0
Interrupt #4	1

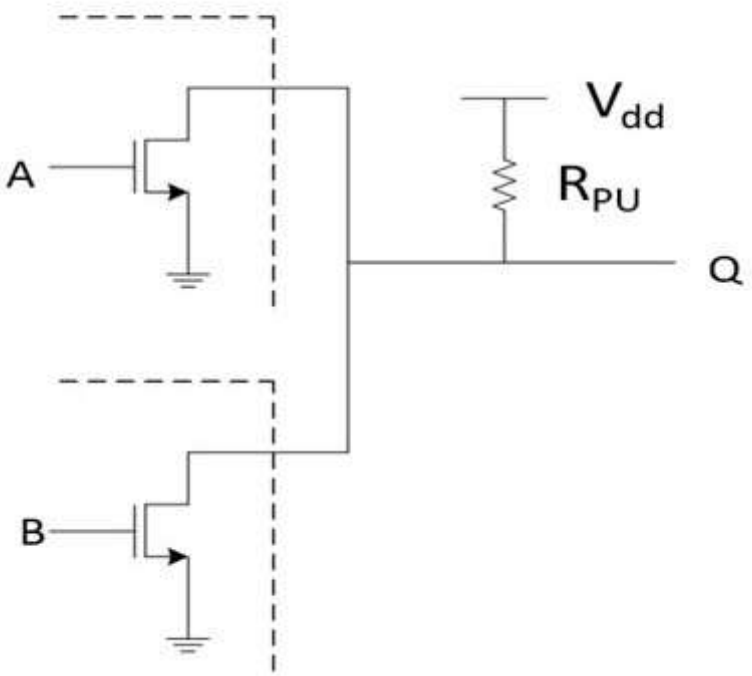
At time t=0, Interrupt #1 and Interrupt #4 are triggered at the same time. Moments later, just before time $t=t_0$, one ISR begins to run.

What is the status of each interrupt at $t=t_0$?

Interrupt	Status
Interrupt #1	[Select]
Interrupt #2	[Select]
Interrupt #3	[Select]
Interrupt #4	[Select]

Question 7 1 pts

The below figure shows an example of Open-Drain Logic (the CMOS equivalent of Open Collector Logic) used in I^2C. If A = 1 (logic high) and B = 0 (logic low), what is the value of Q?



○
0

○
1

○
We cannot tell from this figure.

Question 8 1 pts

An SPI master is connected to a single SPI slave. They contain the following bytes of data in their shift registers:

Component	Data (in binary)
SPI master	10010100
SPI slave	00111011

The SPI master initiates communication with the SPI slave, and the bits are shifted **MSB first**. After ***three*** clock cycles, what is value of the data contained in each shift register? Give your answer in binary ***without*** the 0b prefix.

Component	Data (in binary)
SPI master	
SPI slave	

Question 9 1 pts

The following code uses [?] to represent a bitwise operator in C language. Which bitwise operator should replace [?} to make the statement **TRUE**? Note that the **"0b" prefix means that the value is in binary**. Assume that the DriverLib BIT macros are being used when possible.

```
0b1100 [?] BIT0 == 0b1101
```

- ○ &
- ○ |
- ○ ~

Question 10 1 pts

Consider the following snippet of code.

```
unsigned int array[ 5 ] = {0x343, 0x454, 0x787, 0x989, 0x686};
unsigned int *arrayptr = array + 1;
unsigned int x = *arrayptr;
```

We know the array's location in the memory is at address 0x60002020. Determine the values of arrayptr and x respectively right after completing the above lines of code.

arrayptr: [ ] (Give your answer in hex with 0x prefix)

x: [ ] (Give your answer in hex with 0x prefix)

Question 11 1 pts

Consider a Schmitt trigger with two thresholds: one at 2.4V and another at 2.6V. Also, assume that logic 0 is 0V and logic 1 is 3.3V. Consider a scenario where:

1. the input voltage is 0.7V for several seconds and then
2. the input increases to 1V for several seconds and then
3. the input increases to 2.5V for several seconds and then
4. the input increases to 3V again for several seconds

What is the output **logic level** of the Schmitt trigger at the end of each event described above?

At the end of event 1) [Select]

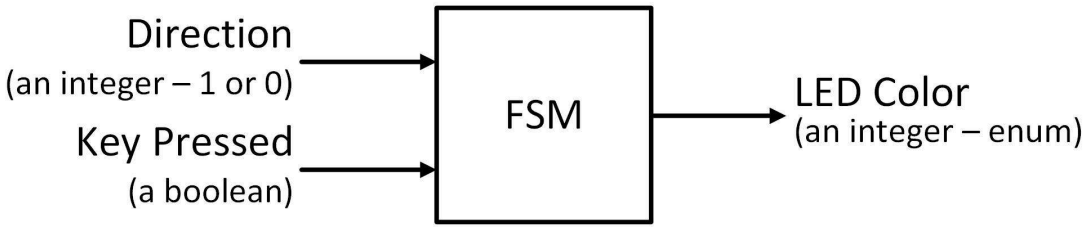
At the end of event 2) [Select]

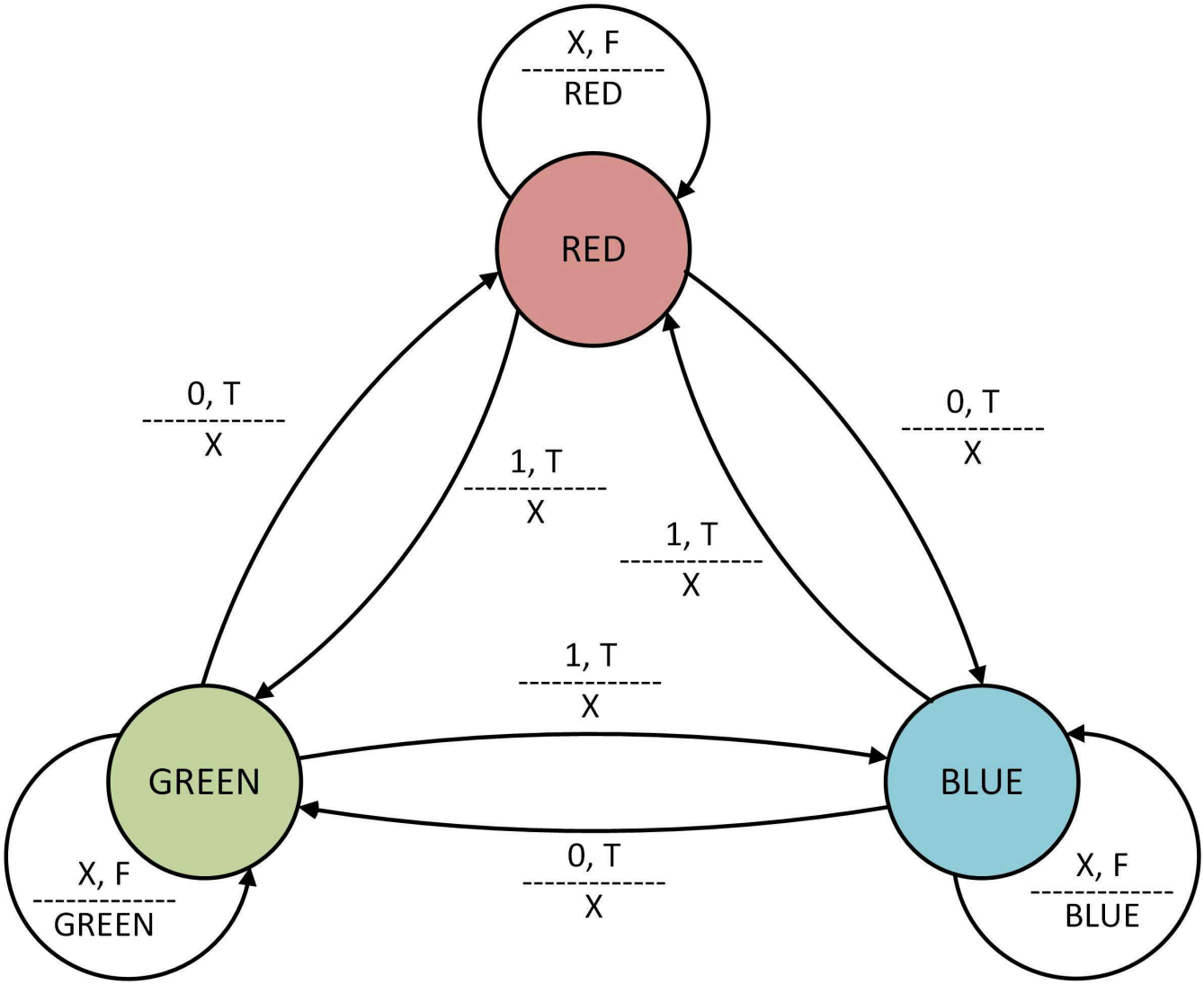
At the end of event 3) [Select]

At the end of event 4) [Select]

Question 12 1 pts

Study the FSM diagram below. The FSM starts on the ***BLUE*** state.





Note that:

- 'X' means "don't care"
- 'T' means "true"
- 'F' means "false"

From the initial state, the user inputs the following input sequence. In which state does the FSM end up?

[Select]

(1, F) -> (0, F) -> (1, F) -> (0, T)

From the initial state, the user inputs the following input sequence. In which state does the FSM end up?

[Select]

(0, T) -> (0, F) -> (1, T) -> (1, T) -> (1, F) -> (1, T)

From the initial state, the user inputs the following input sequence. In which state does the FSM end up?

[Select]

(0, T) -> (1, F) -> (0, T) -> (1, T) -> (0, F) -> (0, T) -> (1, F)

Question 13 1 pts

Suppose you are designing a **Moore Machine** FSM that has exactly one input and one output.

The input has exactly *four* possible values, and the output has exactly *three* possible values.

What is the ***maximum*** number of transition arcs/edges that can emerge from **any individual state** in this FSM?

Question 14 1 pts

If N, the division factor for a UART setup, is calculated to be 96.13, according to Table 24-4, which modulation pattern should we use when sampling the UART bits?

97, 96, 97, 97, 96, 97, 96, 97

96, 96, 96, 97, 96, 96, 96, 96

96, 97, 96, 96, 97, 96, 97, 96

97, 97, 97, 96, 97, 97, 97, 97

Question 15 1 pts

In a UART setup: 1 start bit, 8 data bits, 2 stop bits, ODD parity, Baudrate 38400, **LSB first**, BRCLK is 386 kHz.

What is the data frame for sending the ASCII character ‘S’ in binary **excluding start bit and stop bit**? Your answer should be in a binary format without the 0b prefix.

Question 16 1 pts

Consider a periodic timer in **count-down** mode where the Load value is 10,999. The counter values at the time of E1 (first event) and E2 (second event) are 9000 and 5000, respectively, and four rollovers have happened between the two events. How many counter ***cycles*** have elapsed between the two events?

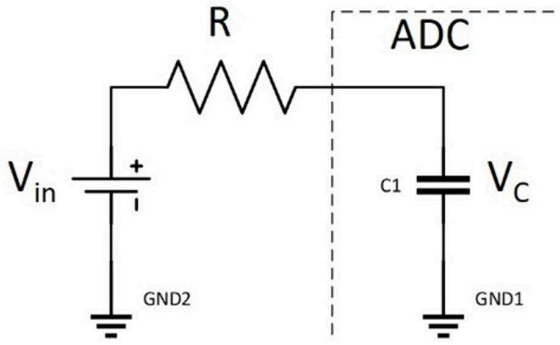
Question 17 1 pts

Assume the clock source of the Timer32 is running at 40MHz. Also, assume the load value is 16,999, the prescaler is 16, and the timer counts down. We want to use a periodic timer and a global variable that keeps track of the total number of **rollovers** from when the timer starts. We turn on an LED when the rollover count is 44 and the timer counter value is 3500. We then want to turn off the LED **after half a second**.

What will be the ***rollover count value*** when we need to turn off the LED?

Question 18 1 pts

The sample & hold (S/H) function of an ADC can be modeled as a lumped RC network, consisting of voltage to be measured (Vin), and a capacitance (Vc) inside the ADC, across which Vin is reflected and measured, as shown in the below figure. Assuming a 14-bit ADC, calculate the maximum theoretical ADC sampling frequency (fs) given R = 2.2K Ω and C_1 = 150 pF **in KHz.** (round down with no decimal points)



Question 19 1 pts

Consider an A/D with n = 3 and Vref- = 0V and Vref+ = 9V. Assume the A/D uses the floor of the conversion formula when turning non-integers to integers. For answers in binary, **use leading zeroes**.

1. How many possible digital outputs are there?
2. What is the A/D analog span? (V)
3. What is the voltage value of the LSB? (Do not round) (V)
4. If the input voltage is 2.55V, what is the output in binary?
5. If the input voltage is 6.92V, what is the output in binary?
6. If the input voltage is 4.3V, what is the output in binary?

Question 20 1 pts

Consider a D/A with n = 4, Vref- = -2V and Vref+ = +3V. For each answer, **do not round**.

1. What is the digital resolution or number of quantization levels?
2. What is the analog span?
3. What is the voltage value of the LSB? (Do not round)
4. If the digital input is 0100, what is the output voltage?
5. If the digital input is 1100, what is the output voltage?
6. If the digital input is 0111, what is the output voltage?

Question 21 1 pts

A function named `uartSendChar` sends a single character over UART and waits for the transmission to complete before returning control to the caller. The function checks the transmit buffer flag until it is clear to ensure the character has been fully sent before exiting.

What type of UART communication does this function implement?

- Blocking
- Non-Blocking

Question 22 1 pts

Consider a scenario where you are programming an MSP432 microcontroller to handle sensor data processing while maintaining responsiveness to user input. Which of the following code snippets best demonstrates an interrupt-based non-blocking approach to reading a sensor value?

Option 1:

```
int readSensor() {
    while (!Sensor_Ready());
    return Get_Sensor_Data();
}
```

Option 2:

```
int readSensor() {
    if (Sensor_Ready()) {
        return Get_Sensor_Data();
    }
    return -1;
}
```

Option 3:

```
void setupSensorCallback() {
    Configure_Sensor_Interrupt();
}

void Sensor_ISR() {
    Sensor_Data = Get_Sensor_Data();
}
```

Option 4:

```
int readSensor() {
    for (int i = 0; i < 1000; i++) {
        if (Sensor_Ready()) {
            return Get_Sensor_Data();
        }
    }
    return -1;
}
```

Option 1

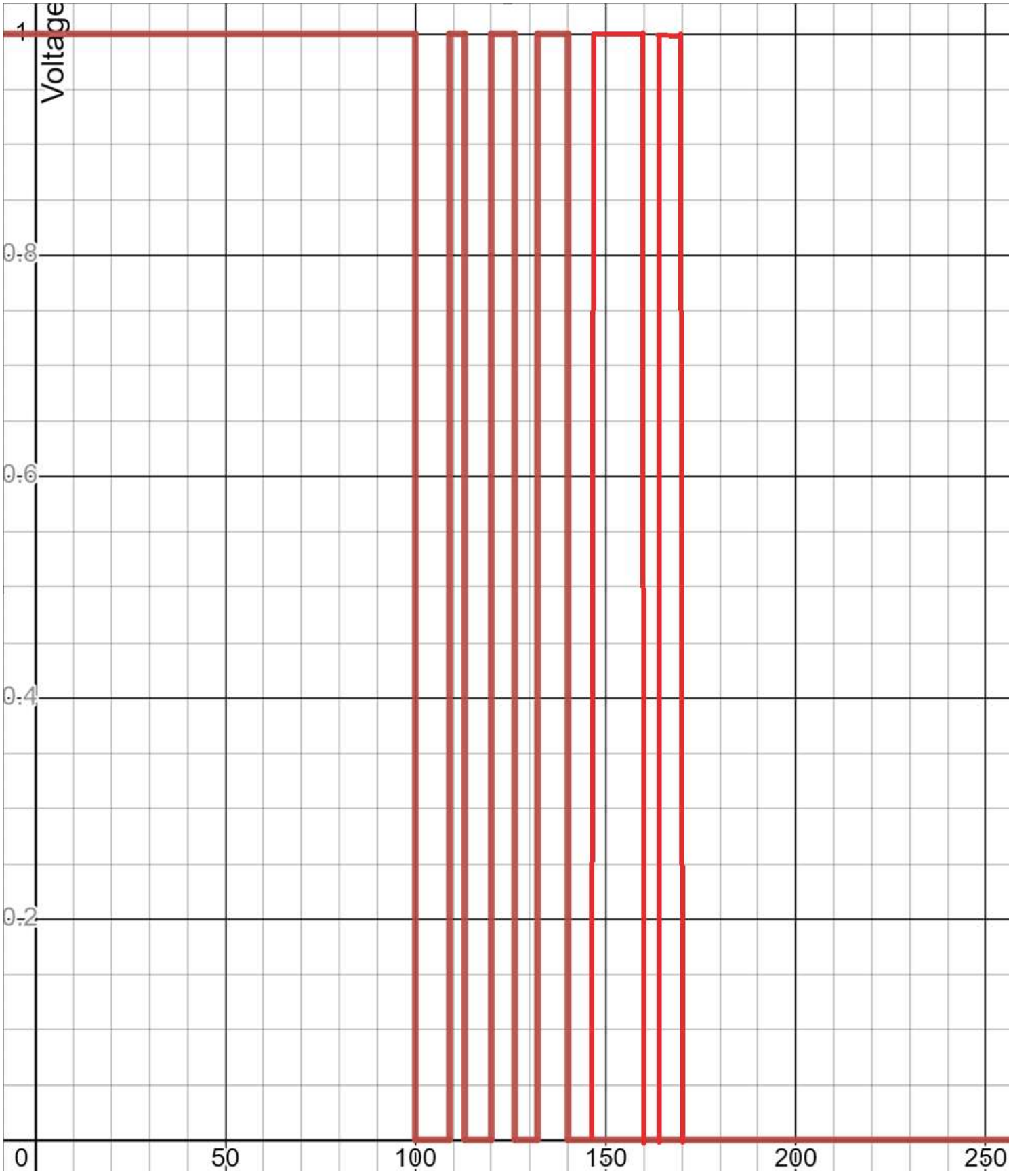
Option 2

Option 3

Option 4

Question 23 1 pts

The figure below is an oscilloscope trace of a button, with the time axis in milliseconds.



Suppose we are making a debouncing FSM that starts a timer after the first falling edge, waits for the timer to expire, and then samples the input.

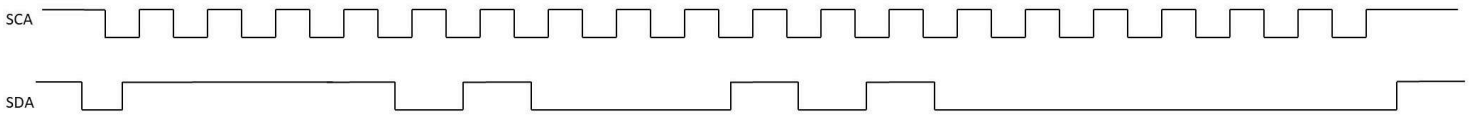
What is the ***minimum*** duration that we should set the timer to (in milliseconds)?

Question 24 1 pts

What is the function used for preventing the watchdog timer from resetting the system? Please write the function call ***exactly as how you would write it in CCS***. Include a **semicolon** and **do not use spaces.**

Question 25 1 pts

Consider the below timing diagram for SCA and SDA on an I2C bus.



Answer these questions:

1. Who creates the start condition? [Select]
2. Who decides if the transfer is a read or write? [Select]
3. What is the address of the slave? [Select]
4. Who sends out the address in the above example? [Select]
5. Who sends out the data in the above example? [Select]

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